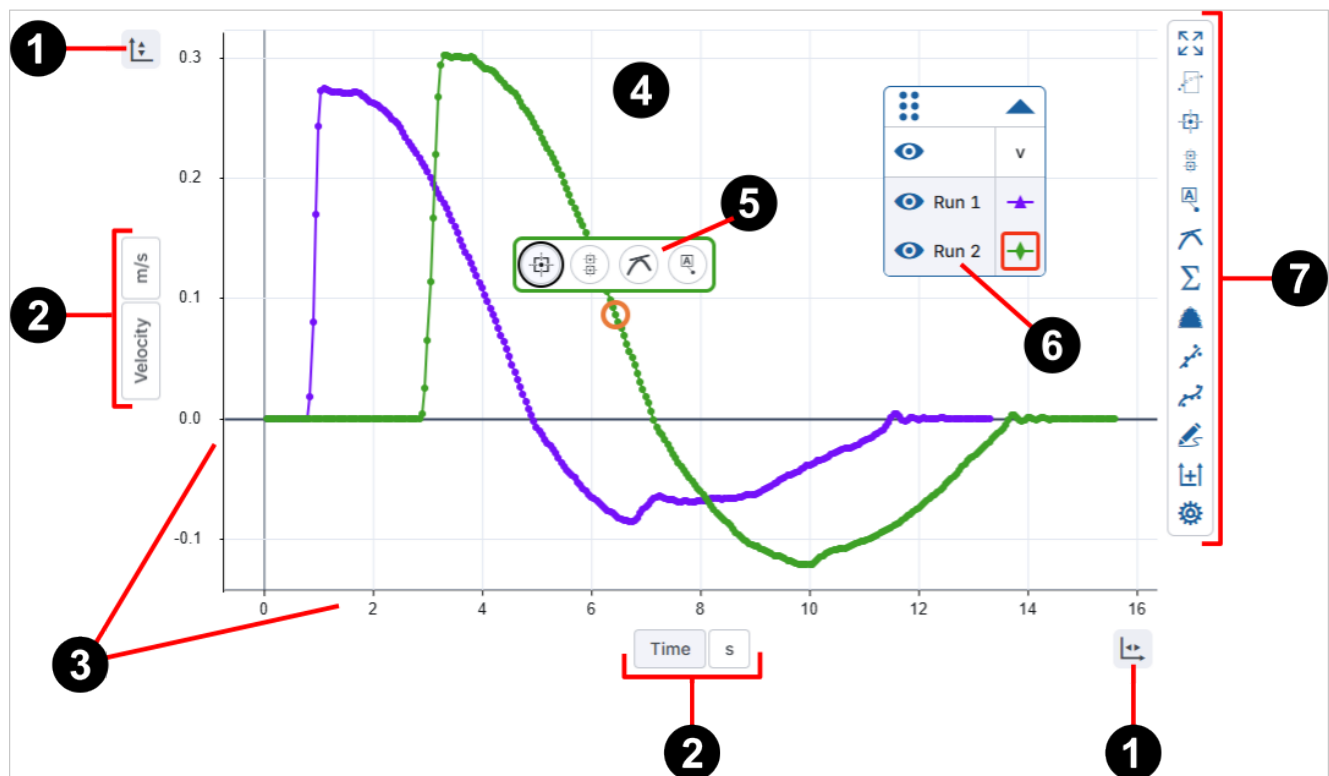




Using SPARKvue

Recording a run, and reading numbers back off the graph.



Screenshot and toolbar icons courtesy of PASCO scientific.

1	Axis range	Set an axis by hand, and lock it so it stops moving.
2	Measurement selectors	Click these to change what the axis shows. The axes are buttons.
3	Axes	Drag an axis to stretch or squash just that one.
4	Plot area	Drag inside the graph to slide the view around.
5	Contextual tools	Appear when you click a point or drag a box round some data.
6	Legend	Click a run here to choose which one you are analysing. The eye hides a run.
7	Graph tools	The toolbar down the right. The ones you need are on the next page.

Getting to a graph

1. Connect the cart, then on **Sensor Data** tick **Position** and **Velocity**. Nothing else — six lines on one graph is unreadable.
2. Under **Templates** choose **Graph**. Your measurements go up the side, time along the bottom.
3. **Start** records, **Stop** ends it. Everything in between is one **run**, and SPARKvue keeps them all so you can compare.

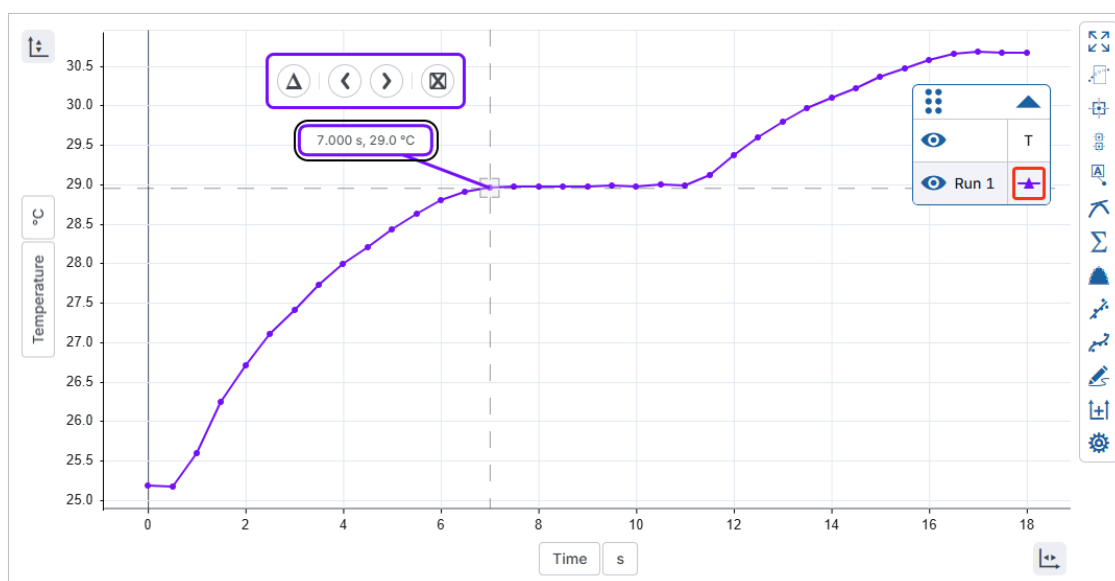
Three habits. Press Start *before* the motion and Stop *after* it. Keep runs to four or five seconds. If a run goes wrong, do it again — recording is free.

The tools you need

	Scale to Fit	Scales the axes so your data fills the screen. Press this first, every time. Most graphs that look wrong are just badly scaled.
	Coordinates	Puts a cursor on the line and reads out the exact values at that point. This is how you get a number off a graph instead of guessing.
	Multi-Coordinate	The same, across several runs at once — “what was the velocity at 2 seconds in <i>both</i> runs?”
	Highlight	Drag a box round part of the graph. The other tools then work on just that stretch.
	Statistics	Minimum, maximum, mean and count for a range of data. Maximum is the useful one: the fastest the cart got.
	Annotation	Sticks a short note on a data point. Good for marking where you let go.

Reading a value off the line

 **Scale to Fit**, then  **Coordinates**, then drag the cursor to the moment you want. The readout gives you both numbers.



Choosing which run you are analysing

The selected run has a **red border** in the legend, and that is the run every tool reads. If your numbers look like somebody else's experiment, this is why. The **eye** hides a run without deleting it.

		
		V X
	Run 1	 
	Run 2	 
	Run 3	 

Worked example

Two tools to leave alone — for now.

 **Slope** draws a line touching the curve and gives its gradient. On a velocity–time graph that gradient is the *acceleration* — which is Lesson 5.

 **Area Under Curve** gives the area beneath the line. That area turns out to be the *distance travelled* — also Lesson 5, and one of the neatest results in the topic.

Not because they are hard. Because working out what they mean is the lesson, and the software will spoil it if you let it.

*Screenshots, icons and tool descriptions from PASCO's SPARKvue help (help.pasco.com). See also: **Connecting SPARKvue**.*